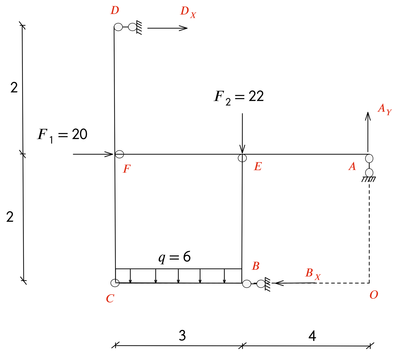


Определяем степень свободы расчётной схемы : $W = 3Д - 2Ш - C_{оп}$

$W = 3 * 3 - 2 * 3 - 3 = 0$, где дисков Д = 3 , шарниров Ш = 3 , опорных стержней $C_{оп} = 3$

Система статически определима и геометрически неизменяема.

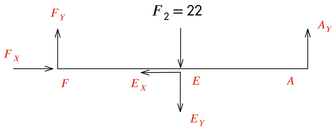
Строим расчётную схему, заменяя связи реакциями связей. Удаляем шарниры , заменяя их реакциями

$$\sum y = 0 \qquad A_y - q * l_1 - F_2 = 0$$
$$A_y = 40\text{кН}$$

$$\sum M_O = 0 \qquad -F_2 * l_2 + F_1 * h_1 - q * l_1 * (0{,}5l_1 + l_2) + D_X * (h_1 + h_2) = 0$$
$$D_X = 36{,}75\text{кН}$$

$$\sum x = 0 \qquad -B_X + F_1 + D_X = 0$$
$$B_X = 56{,}75\text{кН}$$

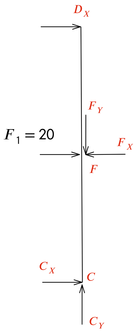
Диск AEF



$$\sum M_E = 0 \qquad -A_Y * l_2 + F_Y * l_1 = 0$$
$$F_Y = 53{,}33\text{кН}$$

$$\sum y = 0 \qquad -F_2 + A_Y + F_Y - E_Y = 0$$
$$E_Y = 71{,}33\text{кН}$$

$$\sum x = 0 \qquad F_X - E_X = 0$$
$$E_X = 93{,}5\text{кН}$$



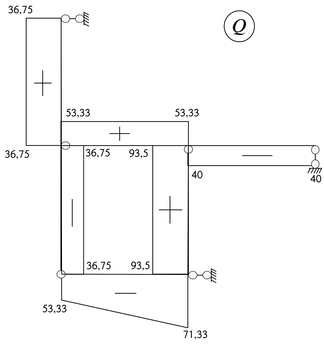
Диск DFC

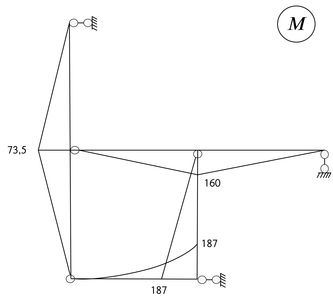
$$\sum M_F = 0 \qquad D_X * h_2 - C_X * h_1 = 0$$
$$C_X = 36{,}75\text{кН}$$

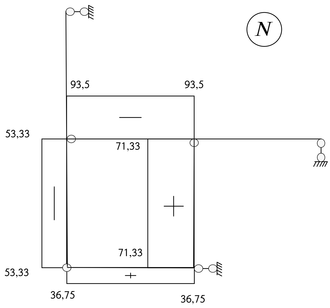
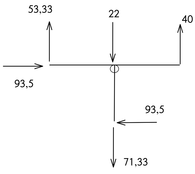
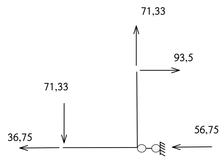
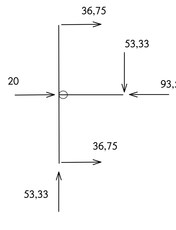
$$\sum x = 0 \qquad D_X + F_1 + C_X - F_X = 0$$
$$F_X = 93{,}5\text{кН}$$

$$\sum y = 0 \qquad -F_Y + C_Y = 0$$
$$C_Y = 53{,}33\text{кН}$$

Строим эпюры







Равновесие узлов